

PROFESSIONAL NURSING COACH

GENERAL SCIENCE

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NN-07 • VERIFIED

Physics + chemistry + biology + environment + scientists • Batches 1–7 • 81 LMR pages + cover & TOC • NORCET / RRB / ESIC / DSSSB

Batch	Units	Pp
B1	Physics-A: units, motion, force, energy, heat	12
B2	Physics-B: transfer, light, sound, circuits start	12
B3	Circuits + Chem-A: series/parallel, atom, table, pH, salts	12
B4	Chem-B + Bio-A: daily chem, fire, cell organelles	12
B5	Bio systems + nutrients: division, systems, vitamins, minerals	11
B6	Disease + plants + eco start: immunity, photo, hormones, chains	11
B7	Environment + scientists: pollution, gases, waste, names	11

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Pp	Block	Focus
4-15	PHY-A (B1)	Units, motion, force, energy, heat
16-27	PHY-B (B2)	Transfer, light, sound, circuits start
28-39	CIR-CHEM (B3)	Series/parallel, atom, table, pH, salts
40-51	CHEM-BIO (B4)	Daily chem, fire, cell organelles
52-62	BIO-SYS (B5)	Division, systems, vitamins, minerals
63-73	DIS-PLANT (B6)	Immunity, photo, hormones, chains
74-84	ENV-SCI (B7)	Pollution, gases, waste, scientists

How to use: formula + unit + trap per page; revise with mocks. Double-QA pending sign-off.

Q	Standard	Status
Q1	All 163 rows mapped, batches 1-7	Done
Q2	No stray XML tags; leak=0 on all batches	Done
Q3	Hybrid style, fonts 10.5-21pt, footers of 11/12	Done

7 base + key derived + symbols + audit! Metre/kg/sec/amp/kelvin/mole/candela + newton/joule/watt/volt/ohm — unit + symbol + audit!

7 base +
symbols + audit

Derived: N/J/W/
V/ohm + audit

Unit + symbol
+ pride!

★ UNIT — MAPS AS

- Base 7: m/kg/s/A/K/mol/cd + symbols + verify.
- Derived: N/J/W/V/ohm + formula link + audit. Ex: N = kg m per s².

⚠ FATAL TRAP

Name-only + 'unit asked' + symbol-miss + formula-miss
+ no-audit = wrong + fail! Unit + symbol + formula.

★ NORCET HIT

HIT: Prefixes + notation next — powers + orders +
audit!

► Unit + symbol + formula + verify + trio?

What + symbol + link + audit.

kilo/milli/micro/nano + powers + order + audit! 10 to power 3/neg 3/neg 6/neg 9 + conversion + audit — prefix + power + audit!

kilo/milli/micro/
nano + power

Notation + order
+ conversion

Prefix + power
+ pride!

▲ PREFIX — CONVERTS AS

- kilo/milli/micro/nano/pico + power + verify.
- Notation + order + conversion + audit. Ex: 3.2×10 to power 5.

⚠ FATAL TRAP

Prefix-only + 'convert' + power-miss + direction-miss + no-audit = wrong + fail! Prefix + power + direction.

★ NORCET HIT

HIT: Scalar/vector + distance next — direction + path + audit!

► Prefix + power + direction + conversion + check?

Value + power + shift + audit.

Magnitude-only vs magnitude+direction + path vs straight + audit! Distance scalar + displacement vector + audit — direction + path + audit!

Scalar vs vector +
direction + audit

Distance vs
displacement + path

Direction + path
+ pride!

◆ QUANTITY — SORTS AS

- Scalar: speed/distance/work/energy + magnitude only.
- Vector: velocity/displacement/force + direction + audit. Ex: displacement zero on return.

⚠ FATAL TRAP

Magnitude-only + 'distance asked' + direction-miss + path-miss = wrong + fail! Direction + path + audit.

★ NORCET HIT

HIT: Speed/velocity + acceleration next — rate + change + audit!

► Scalar + vector + path + direction + sort?

Magnitude + direction + audit.

SPEED + VELOCITY + ACCELERATION

Rate + direction + change-rate + units + audit! $v = s \text{ over } t$ + $a = \text{change } v \text{ over } t$ + audit — rate + direction + audit!

Speed vs velocity +
direction + audit

Acceleration +
units + verify

Rate + change
+ pride!

Σ RATE — MEASURES AS

- Speed scalar + velocity vector + units m per s + verify.
- Acceleration = change in velocity over time + m per s² + audit. Ex: free fall g.

⚠ FATAL TRAP

Value-only + 'fast' + direction-miss + unit-miss + no-audit = wrong + fail! Rate + direction + unit.

★ NORCET HIT

HIT: Graphs + equations next — slope + area + audit!

► Rate + direction + unit + change + check?

v + a + units + audit.

Slope + area + shape + audit! Slope of distance-time = speed + area of velocity-time = displacement + audit — slope + area + audit!

Distance-time slope
= speed + audit

Velocity-time area
= displacement

Slope + area
+ pride!

★ GRAPH — READS AS

- Distance-time: slope = speed + flat = rest + verify.
- Velocity-time: slope = acceleration + area = displacement + speed. Ex: horizontal = uniform.

⚠ FATAL TRAP

Shape-only + 'graph asked' + slope-miss + area-miss = wrong + fail! Slope + area + audit.

★ NORCET HIT

HIT: Equations + force next — formulae + $F=ma$ + audit!

► Slope + area + shape + graph + read?

Slope + area + audit.

$v = u + at$ + $s = ut + \frac{1}{2}at^2$ + $v^2 = u^2 + 2as$ + $F = ma$ + audit! Uniform acceleration + net force + audit — formula + unit + audit!

$v = u + at$ + $s = ut + \frac{1}{2}at^2$ + verify

$v^2 = u^2 + 2as$ +
 $F = ma$ + audit

Formula + unit
+ pride!

▲ EQUATION — SOLVES AS

- Motion: $v = u + at$ + $s = ut + \frac{1}{2}at^2$ + $v^2 = u^2 + 2as$ + verify.
- Force: $F = ma$ + newton + net + direction + audit. Ex: double m = double F .

⚠ FATAL TRAP

Formula-only + 'plug values' + unit-miss + net-miss = wrong + fail! Formula + unit + net + audit.

★ NORCET HIT

HIT: Newton laws + momentum next — inertia + action + audit!

► Formula + unit + net + solve + check?

Pick + plug + unit + audit.

Inertia + $F = ma$ + examples + audit! Rest/motion stays + unbalanced changes + mass resists + audit — inertia + net + audit!

**First law + inertia
 + examples + audit**

**Second law $F=ma$
 + net + units**

**Inertia + net
 + pride!**

◆ LAW — STATES AS

- First: inertia + rest/motion continues + unbalanced changes + verify. Ex: seat belt.
- Second: $F = m a$ + net + direction + audit. Ex: heavier needs bigger F .

⚠ FATAL TRAP

Statement-only + 'law asked' + example-miss + net-miss = wrong + fail! Statement + example + net.

★ NORCET HIT

HIT: Third law + momentum next — action + impulse + audit!

► Law + example + net + inertia + check?

State + example + audit.

Action-reaction + $p = mv$ + impulse + audit! Equal opposite + pairs + momentum conserved + audit — pair + product + audit!

Third law + pairs
+ examples + audit

Momentum $p=mv$ +
impulse + verify

Pair + product
+ pride!

Σ PAIR — ACTS AS

- Third: equal opposite + different bodies + verify. Ex: rocket push.
- Momentum $p = m v$ + impulse = $F t$ + conserved + audit. Ex: recoil.

⚠ FATAL TRAP

Single-body + 'reaction' + pair-miss + direction-miss = wrong + fail! Pair + direction + audit.

★ NORCET HIT

HIT: Work + power next — $F d$ + W over t + audit!

► Pair + product + direction + conserved + check?

Action + p + audit.

WORK + POWER

$W = F d + P = W \text{ over } t + \text{units} + \text{audit!}$ Force along displacement + joule + watt + audit — $F d + \text{rate} + \text{audit!}$

**Work $W = F d +$
 joule + verify**

**Power $P = W \text{ over } t$
 + watt + audit**

**$F d + \text{rate}$
 + pride!**

★ WORK — COUNTS AS

- Work = $F \times d$ along force + joule + verify. Ex: zero when perpendicular.
- Power = work over time + watt + audit. Ex: 1 HP link.

⚠ FATAL TRAP

Value-only + 'work asked' + angle-miss + unit-miss = wrong + fail! Angle + unit + audit.

★ NORCET HIT

HIT: KE + PE + conservation next — half $m v^2 + m g h$ + audit!

► $F d + \text{rate} + \text{unit} + \text{angle} + \text{check?}$

$W + P + \text{units} + \text{audit}.$

KINETIC + POTENTIAL ENERGY

KE = half m v² + PE = m g h + units + audit! Motion + height + interconvert + audit — motion + height + audit!

**KE half m v² +
 motion + verify**

**PE m g h + height
 + audit**

**Motion + height
 + pride!**

▲ ENERGY — STORES AS

- KE = half m v squared + speed matters squared + verify. Ex: double v = 4x KE.
- PE = m g h + height + audit. Ex: raised water + wound spring.

⚠ FATAL TRAP

Formula-only + 'energy asked' + squared-miss + unit-miss = wrong + fail! Squared + unit + audit.

★ NORCET HIT

HIT: Conservation + machines next — total + advantage + audit!

► Motion + height + squared + unit + check?

KE + PE + audit.

Total constant + MA = load over effort + audit! Energy converts + machines multiply force + audit — total + ratio + audit!

**Conservation +
 convert + verify**

**MA load over effort
 + examples + audit**

**Total + ratio
 + pride!**

◆ SYSTEM — KEEPS AS

- Conservation: total same + KE/PE swap + verify. Ex: pendulum + falling body.
- Machines: MA = load over effort + lever/pulley + audit. Ex: longer effort arm wins.

⚠ FATAL TRAP

Loss-only + 'energy lost' + convert-miss + ratio-miss = wrong + fail! Convert + ratio + audit.

★ NORCET HIT

HIT: Heat + temperature next — energy vs measure + audit!

► Total + convert + ratio + system + check?

Total + MA + audit.

Energy vs measure + C/F/K + convert + audit! Heat joule + temp degree + ice/steam points + audit — energy vs degree + audit!

Heat vs temp +
units + verify

C/F/K + convert
+ points + audit

Energy vs degree
+ pride!

Σ MEASURE — READS AS

- Heat = energy flow + joule + hot to cold + verify.
- Temp scales + convert + ice/steam points + audit. Ex: 0 C = 32 F = 273 K.

⚠ FATAL TRAP

Same-only + 'heat asked' + unit-miss + convert-miss = wrong + fail! Unit + convert + audit.

★ NORCET HIT

HIT: Till end: transfer + light + sound + audit!

► Energy + degree + convert + points + check?

Heat + C/F/K + audit.

Solid-touch vs fluid-flow + examples + audit! Metal rod + sea breeze + audit — medium + flow + audit!

**Conduction solid/
touch + examples**

**Convection fluid/
flow + examples**

**Medium + flow
+ pride!**

★ TRANSFER — MOVES BY

- Conduction: solids + touch + metals best + verify. Ex: spoon heats.
- Convection: fluids + bulk flow + audit. Ex: sea/land breeze + radiator.

⚠ FATAL TRAP

Mode-only + 'heat moves' + medium-miss + example-miss = wrong + fail! Medium + example + audit.

★ NORCET HIT

HIT: Radiation + expansion next — waves + spacing + audit!

► Medium + flow + example + transfer + sort?

Solid vs fluid + audit.

No-medium waves + solids/liquids/gases expand + audit! Sun warmth + gaps in rails + audit — waves + gaps + audit!

Radiation no-medium
+ waves + verify

Expansion + gaps/
joints + audit

Waves + gaps
+ pride!

▲ EFFECT — SHOWS AS

- Radiation: no medium + dark absorbs + verify. Ex: sun + cooker.
- Expansion: heat expands + gaps/joints + audit. Ex: rails + bridges + bimetal.

⚠ FATAL TRAP

Heat-only + 'expands' + medium-miss + direction-miss
= wrong + fail! Medium + gap + audit.

★ NORCET HIT

HIT: Specific heat + change of state next — capacity + latent + audit!

► Waves + gaps + medium + effect + check?

Radiation + expand + audit.

SPECIFIC HEAT + LATENT HEAT

Capacity + hidden heat + water high + audit! $Q = m c \text{ change } T + \text{melt/boil flat} + \text{audit}$ — capacity + flat + audit!

Specific heat +
water high + verify

Latent melt/boil
flat + audit

Capacity + flat
+ pride!

◆ HEAT — ABSORBS AS

- Specific heat: $Q = m c \text{ change } T$ + water highest + verify. Ex: sea moderates.
- Latent: flat at melt/boil + hidden + audit. Ex: steam burn worse.

⚠ FATAL TRAP

Temp-only + 'heats' + flat-miss + capacity-miss = wrong
+ fail! Flat + capacity + audit.

★ NORCET HIT

HIT: Reflection + mirrors next — laws + image + audit!

► Capacity + flat + water + heat + check?

c + latent + audit.

Laws + virtual image + lateral + audit! Angle in = angle out + same distance behind + audit — angle + image + audit!

**Laws angle in =
 angle out + verify**

**Plane virtual + same
 size/distance + audit**

**Angle + image
 + pride!**

Σ MIRROR — IMAGES AS

- Laws: angle in = angle out + same plane + verify.
- Plane: virtual + erect + same size/distance + lateral flip + audit. Ex: periscope pair.

⚠ FATAL TRAP

Real-only + 'image asked' + virtual-miss + flip-miss = wrong + fail! Virtual + flip + audit.

★ NORCET HIT

HIT: Spherical + refraction next — concave/convex + bend + audit!

► Angle + virtual + flip + mirror + check?

Laws + image + audit.

Concave/convex + bend at boundary + audit! Converge/diverge + rarer-denser + audit — curve + bend + audit!

Concave/convex +
 uses + verify

Refraction bend +
 rarer-denser + audit

Curve + bend
 + pride!

★ BEND — FORMS AS

- Concave converges + convex diverges + verify. Ex: torch + rear view.
- Refraction: speed changes + bends + audit. Ex: straw bent + pool shallow.

⚠ FATAL TRAP

Mirror-only + 'bend asked' + converge-miss + medium-miss = wrong + fail! Converge + medium + audit.

★ NORCET HIT

HIT: Lenses + eye next — convex/concave + retina + audit!

► Curve + medium + bend + use + check?

Mirror + refract + audit.

Convex/concave + retina/lens/ciliary + audit! Converge/diverge + real image on retina + audit — lens + retina + audit!

Convex/concave +
uses + verify

Eye lens/retinal/
ciliary + audit

Lens + retina
+ pride!

▲ LENS — FOCUSES AS

- Convex converges + concave diverges + verify. Ex: magnifier + specs.
- Eye: lens focuses + retina senses + ciliary adjusts + audit. Ex: real inverted image.

⚠ FATAL TRAP

Glass-only + 'lens asked' + converge-miss + retina-miss = wrong + fail! Converge + retina + audit.

★ NORCET HIT

HIT: Defects + instruments next — myopia + scope + audit!

► Converge + retina + adjust + focus + check?

Lens + eye + audit.

Myopia/hypermetropia + concave/convex fix + scope + audit! Near/far blur + lens corrects + audit — defect + fix + audit!

**Myopia concave +
hyper convex + verify**

**Microscope/telescope
+ magnify + audit**

**Defect + fix
+ pride!**

◆ DEFECT — FIXES AS

- Myopia: far blur + concave + verify. Hyper: near blur + convex + audit.
- Instruments: magnify + lenses/mirrors + audit. Ex: lab + stars.

⚠ FATAL TRAP

Blur-only + 'vision asked' + lens-miss + near/far-miss = wrong + fail! Near/far + lens + audit.

★ NORCET HIT

HIT: Sound + frequency next — wave + pitch + audit!

► Near + far + lens + defect + fix?

Defect + lens + audit.

Mechanical wave + f + pitch link + audit! Needs medium + faster in solids + audit — medium + pitch + audit!

Sound mechanical +
needs medium + verify

Frequency + pitch
+ link + audit

Medium + pitch
+ pride!

Σ WAVE — TRAVELS AS

- Sound: mechanical + medium must + verify. Ex: no sound in vacuum.
- Frequency high = shrill + low = grave + audit. Ex: woman vs man voice.

⚠ FATAL TRAP

Air-only + 'sound asked' + medium-miss + pitch-miss =
wrong + fail! Medium + pitch + audit.

★ NORCET HIT

HIT: Loudness + wave relation next — amplitude + $v = f \lambda$
+ audit!

► Medium + pitch + wave + sound + check?

Medium + f + audit.

Amplitude + $v = f \lambda$ + audit! Bigger swing = louder + speed fixed in medium + audit — swing + product + audit!

**Amplitude + loudness
+ link + verify**

**$v = f \lambda$ +
product + audit**

**Swing + product
+ pride!**

★ RELATION — LINKS AS

- Amplitude big = loud + small = soft + verify. Ex: whisper vs shout.
- $v = f \times \lambda$ + units + audit. Ex: double f = half λ .

⚠ FATAL TRAP

Loud-only + 'wave asked' + amplitude-miss + product-miss = wrong + fail! Amplitude + product + audit.

★ NORCET HIT

HIT: Ranges + ultrasound next — audible + uses + audit!

► Swing + product + units + wave + check?

Amplitude + $v=f \lambda$ + audit.

Infrasonic/audible/ultrasonic + scan/clean + audit! 20 to 20000 Hz human + bats/dogs beyond + audit — range + use + audit!

Ranges + 20-20000
Hz + verify

Ultrasound scan/
clean + uses + audit

Range + use
+ pride!

▲ RANGE — HEARS AS

- Infrasonic below 20 + audible 20-20000 + ultrasonic above + verify.
- Ultrasound: scan/clean/sonar + audit. Ex: pregnancy scan + ships.

⚠ FATAL TRAP

Hear-only + 'sound asked' + range-miss + use-miss =
wrong + fail! Range + use + audit.

★ NORCET HIT

HIT: Echo + charge/current next — reflect + $Q = It$ +
audit!

► Range + use + Hz + scan + check?

Range + uses + audit.

Reflect + $Q = I t$ + audit! 17 m min for echo + coulomb + audit — reflect + flow + audit!

Echo reflect +
17 m + verify

Charge $Q=It$ +
current + audit

Reflect + flow
+ pride!

◆ SIGNAL — RETURNS AS

- Echo: reflection + 17 m min + verify. Ex: sonar + hills.
- Charge $Q = I \times t$ + coulomb + current flow + audit. Ex: battery drain.

⚠ FATAL TRAP

Sound-only + 'echo asked' + distance-miss + flow-miss
= wrong + fail! Distance + flow + audit.

★ NORCET HIT

HIT: Voltage + Ohm law next — push + $V = IR$ + audit!

► Reflect + flow + distance + charge + check?

Echo + $Q=It$ + audit.

Push + $V = IR$ + audit! Potential difference volt + resistance ohm + audit — push + ratio + audit!

**Voltage push +
volt + verify**

**$V=IR$ + ohm +
ratio + audit**

**Push + ratio
+ pride!**

Σ CIRCUIT — DRIVES AS

- Voltage: push + volt + verify. Ex: cell + mains.
- Ohm: $V = IR$ + ohm + audit. Ex: double $V =$ double I .

⚠ FATAL TRAP

Current-only + 'circuit asked' + push-miss + ratio-miss = wrong + fail! Push + ratio + audit.

★ NORCET HIT

HIT: Till end: series/parallel + power + audit!

► Push + ratio + volt + ohm + check?

$V + IR$ + audit.

SERIES + PARALLEL RESISTORS

Add vs reciprocal + same I vs same V + audit! $R_s = \text{sum} + 1 \text{ over } R_p = \text{sum reciprocals} + \text{audit}$ —
add vs share + audit!

Series $R_s = \text{sum} +$
same I + verify

Parallel $1/R_p = \text{sum}$
same V + audit

Add vs share
+ pride!

★ COMBO — JOINS AS

- Series: $R_s = R_1 + R_2 + \text{same current} + \text{verify}$. Ex: lights in a row.
- Parallel: same voltage + $1 \text{ over } R_p = \text{sum} + \text{audit}$. Ex: home wiring.

⚠ FATAL TRAP

Add-all + 'resistors asked' + mode-miss + same-miss =
wrong + fail! Mode + same + audit.

★ NORCET HIT

HIT: Power + energy + safety next — watt + kWh + fuse
+ audit!

► Mode + same + sum + combo + check?

Series vs parallel + audit.

P = V I + kWh commercial + audit! Watt + Board unit + bill math + audit — VI + kWh + audit!

**Power $P=VI$ +
watt + verify**

**Energy kWh + bill
+ math + audit**

**VI + kWh
+ pride!**

▲ BILL — COUNTS AS

- Power $P = V \times I = I^2 R$ + watt + verify. Ex: 100 W bulb.
- Energy kWh = units + bill = units x rate + audit. Ex: 2 kW x 3 h = 6 units.

⚠ FATAL TRAP

Watt-only + 'power asked' + formula-miss + unit-miss =
wrong + fail! Formula + kWh + audit.

★ NORCET HIT

HIT: Conductors + magnets + safety next — flow +
poles + fuse + audit!

► VI + kWh + rate + bill + math?

P + units + audit.

CONDUCTORS + MAGNETS + ELECTROMAGNET

Flow vs block + poles + coil-current + audit! Copper flows + N/S attract + soft-iron core + audit — flow + poles + audit!

Conductors/insulators
+ examples + verify

Magnet poles +
electromagnet + audit

Flow + poles
+ pride!

◆ FIELD — PULLS AS

- Conductors flow + insulators block + verify. Ex: copper vs rubber.
- Magnet N/S + like repels + electromagnet = coil + current + soft core + audit. Ex: crane + bell.

⚠ FATAL TRAP

Iron-only + 'magnet asked' + pole-miss + current-miss = wrong + fail! Poles + current + audit.

★ NORCET HIT

HIT: Safety + matter next — fuse + states + audit!

► Flow + poles + current + field + sort?

Conductor + magnet + audit.

DOMESTIC SAFETY + STATES OF MATTER

Fuse/MCB/earth + solid/liquid/gas + audit! Overload melts + shape/volume rules + audit — protect + shape + audit!

Fuse/MCB/earth +
protect + verify

Solid/liquid/gas +
shape/volume + audit

Protect + shape
+ pride!

Σ HOME — STAYS SAFE AS

- Safety: fuse melts + MCB trips + earth guards + verify. Ex: 3-pin plug.
- States: solid fixed + liquid flows + gas fills + audit. Ex: ice/water/steam.

⚠ FATAL TRAP

Wire-only + 'safety asked' + device-miss + state-miss = wrong + fail! Device + state + audit.

★ NORCET HIT

HIT: Changes + physical/chemical next — melt vs burn + audit!

► Protect + shape + device + state + check?

Fuse + states + audit.

CHANGES OF STATE + PHYSICAL/CHEMICAL

Melt/freeze/boil/condense + reversible vs new-substance + audit! Heat in/out + rust/burn new + audit — reverse vs new + audit!

Melt/freeze/boil +
heat + verify

Physical reverse vs
chemical new + audit

Reverse vs new
+ pride!

★ CHANGE — SORTS AS

- States change: melt/freeze/boil/condense/sublime + verify. Ex: camphor sublimates.
- Physical reversible + chemical new substance + audit. Ex: melt vs burn/rust.

⚠ FATAL TRAP

Change-only + 'change asked' + reverse-miss +
new-miss = wrong + fail! Reverse + new + audit.

★ NORCET HIT

HIT: Atom + element/compound next — particle + pure
+ audit!

► Reverse + new + heat + change + sort?

Physical vs chemical + audit.

ATOM + ELEMENT/COMPOUND/MIXTURE

Particle + pure vs combined vs mixed + audit! Atom/molecule/ion + element/compound/mixture + audit — particle + pure + audit!

Atom/molecule/ion
+ particle + verify

Element/compound/
mixture + pure + audit

Particle + pure
+ pride!

▲ MATTER — BUILDS AS

- Atom smallest + molecule bonded + ion charged + verify. Ex: Na⁺ / Cl⁻.
- Element pure-one + compound fixed-ratio + mixture separable + audit. Ex: air mixture.

⚠ FATAL TRAP

Mix-only + 'matter asked' + ratio-miss + separable-miss
= wrong + fail! Ratio + separable + audit.

★ NORCET HIT

HIT: Mixtures + subatomic next — uniform + p/n/e +
audit!

► Particle + ratio + separable + matter + sort?

Atom + element + audit.

Uniform vs non-uniform + p/n/e + audit! Solution/alloy vs sand + proton/neutron/electron + audit
 — uniform + charge + audit!

Homogeneous vs
 hetero + verify

Proton/neutron/
 electron + charge

Uniform + charge
 + pride!

◆ TINY — SPLITS AS

- Homogeneous uniform + hetero non-uniform + verify. Ex: salt solution vs sand.
- Proton + / neutron 0 / electron - + mass + audit. Ex: e lightest.

⚠ FATAL TRAP

Mix-only + 'tiny asked' + uniform-miss + charge-miss =
 wrong + fail! Uniform + charge + audit.

★ NORCET HIT

HIT: Atomic numbers + isotopes next — Z/A + same-Z
 + audit!

► Uniform + charge + mass + tiny + sort?

Mixture + p/n/e + audit.

ATOMIC NUMBER + ISOTOPES

Z = protons + A = p + n + same-Z + audit! Identity + mass + uses + audit — Z + same + audit!

Z protons + A p+n
 + identity + verify

Isotopes same-Z +
 uses + audit

Z + same
 + pride!

Σ IDENTITY — FIXES AS

- Z = protons = electrons + A = protons + neutrons + verify. Ex: carbon Z 6.
- Isotopes: same Z + diff neutrons + audit. Ex: C-12/C-14 + U-235 fuel.

⚠ FATAL TRAP

Mass-only + 'atom asked' + Z-miss + same-miss =
 wrong + fail! Z + same + audit.

★ NORCET HIT

HIT: Periodic + acids next — table + sour + audit!

► **Z + A + same + identity + check?**

Z/A + isotopes + audit.

Periods/groups + sour/soapy + audit! Metals-left + nonmetals-right + H+ vs OH- + audit — rows + ions + audit!

Periods/groups + trends + verify	Acid H+ vs base OH- + audit	Rows + ions + pride!
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★ TABLE — ARRANGES AS

- Periods rows + groups columns + metals left + verify. Ex: group 1 alkali.
- Acid sour H+ + base bitter OH- + soapy + audit. Ex: lemon vs soap.

⚠ FATAL TRAP

Name-only + 'table asked' + trend-miss + ion-miss = wrong + fail! Trend + ion + audit.

★ NORCET HIT

HIT: pH + indicators next — scale + colour + audit!

► Rows + ions + sour + table + sort?

Periodic + acid/base + audit.

0-14 + 7 neutral + colour flips + audit! Litmus/phenol/methyl + acid-blue-red + audit — number + colour + audit!

pH 0-14 + 7
neutral + verify

Litmus/phenol/methyl
+ flips + audit

Number + colour
+ pride!

▲ SCALE — READS AS

- pH below 7 acid + 7 neutral + above 7 base + verify. Ex: blood 7.4.
- Litmus red/blue + phenol pink in base + methyl orange + audit. Ex: lab tests.

⚠ FATAL TRAP

Colour-only + 'pH asked' + number-miss + flip-miss = wrong + fail! Number + flip + audit.

★ NORCET HIT

HIT: Neutralization + common acids next — salt-water + names + audit!

► Number + flip + scale + colour + read?

pH + indicator + audit.

Acid + base = salt + water + names + audit! HCl/H₂SO₄/HNO₃ + uses + audit — salt + name + audit!

Neutralize salt +
water + verify

HCl/H₂SO₄/HNO₃ +
uses + audit

Salt + name
+ pride!

◆ REACTION — ENDS AS

- Neutralization: acid + base = salt + water + verify. Ex: antacid relief.
- Common acids: HCl/H₂SO₄/HNO₃ + bases NaOH/CaO + audit. Ex: battery + cleaner.

⚠ FATAL TRAP

Mix-only + 'acid asked' + product-miss + name-miss = wrong + fail! Product + name + audit.

★ NORCET HIT

HIT: Salts + daily-pH next — common + life + audit!

► Product + name + use + acid + check?

Salt-water + names + audit.

COMMON SALTS

NaCl/washing/baking/plaster + uses + audit! Table/soda/gypsum + daily + audit — salt + use + audit!

NaCl/washing/baking
+ names + verify

Plaster/gypsum +
uses + audit

Salt + use
+ pride!

Σ SALT — USES AS

- Table NaCl + washing Na₂CO₃ + baking NaHCO₃ + verify. Ex: kitchen + clean.
- Plaster POP + gypsum + audit. Ex: casts + walls.

⚠ FATAL TRAP

White-only + 'salt asked' + formula-miss + use-miss =
wrong + fail! Formula + use + audit.

★ NORCET HIT

HIT: Till end: daily-pH + metals + audit!

► Formula + use + salt + daily + check?

Salts + uses + audit.

DAILY pH + METAL PROPERTIES

Stomach/soil/rain + lustre/malleable + audit! pH in life + metal traits + audit — life + trait + audit!

**Stomach/soil/rain
+ pH + verify**

**Lustre/malleable/
conduct + audit**

**Life + trait
+ pride!**

★ DAILY — APPLIES AS

- pH life: stomach acid + soil + rain + verify. Ex: antacid + lime.
- Metals: lustre + malleable + ductile + conduct + audit. Ex: gold + copper.

⚠ FATAL TRAP

Lab-only + 'pH asked' + life-miss + trait-miss = wrong + fail! Life + trait + audit.

★ NORCET HIT

HIT: Reactivity + series next — displacement + order + audit!

► Life + trait + pH + metal + apply?

Daily + props + audit.

Tendency + K Na Ca Mg Al Zn Fe + audit! Displacement order + exam list + audit — order + displace + audit!

Reactivity +
tendency + verify

Series K Na Ca Mg
Al Zn Fe + audit

Order + displace
+ pride!

▲ SERIES — ORDERS AS

- Reactivity: ease of reaction + verify. Ex: Na explodes in water.
- Series high-yield: K Na Ca Mg Al Zn Fe Pb H Cu + displaces below + audit. Ex: Zn + CuSO₄.

⚠ FATAL TRAP

Metal-only + 'reacts asked' + order-miss + displace-miss = wrong + fail! Order + displace + audit.

★ NORCET HIT

HIT: Corrosion + prevention next — rust + coat + audit!

► Order + displace + series + exam + check?

Series + displace + audit.

Rust needs air + water + coat/galvanize + audit! Iron oxide + oil/paint/zinc + audit — rust + shield + audit!

Rust air + water
+ iron + verify

Oil/paint/zinc +
shield + audit

Rust + shield
+ pride!

◆ RUST — BLOCKS AS

- Corrosion: iron + air + water = rust + verify. Ex: gates + ships.
- Prevent: oil/paint/galvanize/alloy + audit. Ex: zinc coat + stainless.

⚠ FATAL TRAP

Rust-only + 'corrode asked' + needs-miss + shield-miss
= wrong + fail! Needs + shield + audit.

★ NORCET HIT

HIT: Alloys + uses next — mix + purpose + audit!

► Needs + shield + rust + coat + block?

Rust + prevent + audit.

ALLOYS + METAL/NONMETAL USES

Mix wins + brass/steel + wires/foil + audit! Purpose + examples + audit — mix + use + audit!

Brass/steel +
mix + verify

Wires/foil/tyre +
uses + audit

Mix + use
+ pride!

Σ MIX — WINS AS

- Alloys: brass + bronze + steel + verify. Ex: bells + tools.
- Uses: Cu wires + Al foil + S tyres + graphite + audit. Ex: daily items.

⚠ FATAL TRAP

Pure-only + 'alloy asked' + mix-miss + use-miss =
wrong + fail! Mix + use + audit.

★ NORCET HIT

HIT: Soap + bleach next — clean + whiten + audit!

► Mix + use + alloy + daily + check?

Alloy + uses + audit.

Fat + alkali vs synthetic + whiten + audit! Hard-water fail vs works + chlorine + audit — clean + whiten + audit!

Soap vs detergent
+ hard-water + verify

Bleach chlorine +
whiten + audit

Clean + whiten
+ pride!

★ CLEAN — WASHES AS

- Soap: fat + alkali + fails in hard water + verify. Detergent works + audit.
- Bleach: chlorine + whiten + disinfect + audit. Ex: clothes + water.

⚠ FATAL TRAP

Soap-only + 'clean asked' + hard-miss + agent-miss =
wrong + fail! Hard + agent + audit.

★ NORCET HIT

HIT: Fertilizers + fuels next — NPK + burn + audit!

► **Hard + agent + clean + whiten + sort?**

Soap + bleach + audit.

NPK + burn value + audit! Urea/potash + petrol/diesel/CNG + audit — feed + burn + audit!

**NPK/urea/potash
+ feed + verify**

**Petrol/diesel/CNG +
burn + audit**

**Feed + burn
+ pride!**

▲ FIELD — BURNS AS

- Fertilizers: NPK + urea + potash + verify. Ex: crops need N.
- Fuels: petrol/diesel/CNG/LPG + calorific + audit. Ex: kitchen + vehicles.

⚠ **FATAL TRAP**

Name-only + 'fuel asked' + NPK-miss + burn-miss =
wrong + fail! NPK + burn + audit.

★ **NORCET HIT**

HIT: Polymers + household next — plastic + daily +
audit!

► **Feed + burn + NPK + fuel + sort?**

Fertilizer + fuel + audit.

POLYMERS + HOUSEHOLD CHEMICALS

Repeat units + plastic/rubber + daily + audit! Polythene/PVC + cleaners + audit — repeat + daily + audit!

**Polythene/PVC +
repeat + verify**

**Cleaners/phenyl +
daily + audit**

**Repeat + daily
+ pride!**

◆ PLASTIC — REPEATS AS

- Polymers: repeat units + plastic/rubber/fibre + verify. Ex: bags + tyres.
- Household: cleaners/phenyl/naphthalene + audit. Ex: kitchen + toilet.

⚠ **FATAL TRAP**

Plastic-only + 'polymer asked' + repeat-miss + daily-miss = wrong + fail! Repeat + daily + audit.

★ **NORCET HIT**

HIT: Hard water + fire triangle next — scale + fuel-oxygen-heat + audit!

► **Repeat + daily + plastic + home + sort?**

Polymer + home + audit.

Ca/Mg scale + fuel + oxygen + heat + audit! Soap fails + boil softens + remove one leg + audit — scale + triangle + audit!

Hard Ca/Mg +
scale + verify

Fuel/oxygen/heat +
remove-one + audit

Scale + triangle
+ pride!

Σ SCALE — FIGHTS AS

- Hard: Ca/Mg + soap scum + scale + boil/ion-exchange + verify.
- Fire: fuel + oxygen + heat + remove one + audit. Ex: blanket + CO2.

⚠ FATAL TRAP

Water-only + 'hard asked' + ion-miss + leg-miss =
wrong + fail! Ions + legs + audit.

★ NORCET HIT

HIT: Cell theory + prokaryote next — unit + no-nucleus
+ audit!

► Ions + legs + scale + fire + check?

Hard + triangle + audit.

Unit of life + no-nucleus + audit! Schleiden/Schwann + bacteria + audit — unit + simple + audit!

Theory unit +
life + verify

Prokaryote no-nucleus
+ bacteria + audit

Unit + simple
+ pride!

★ CELL — STARTS AS

- Theory: cell = unit + all from cells + verify. Ex: plants + animals.
- Prokaryote: no nucleus + no organelles + bacteria + audit. Ex: E. coli.

⚠ FATAL TRAP

Cell-only + 'cell asked' + unit-miss + nucleus-miss =
wrong + fail! Unit + nucleus + audit.

★ NORCET HIT

HIT: Plant/animal + membrane next — wall + gate +
audit!

► Unit + nucleus + simple + cell + sort?

Theory + prokaryote + audit.

Wall vs no-wall + gate + audit! Chloroplast/vacuole big vs small + selective + audit — wall + gate + audit!

**Plant vs animal +
wall + verify**

**Membrane/wall +
gate + audit**

**Wall + gate
+ pride!**

▲ GATE — GUARDS AS

- Plant: wall + big vacuole + chloroplast + verify. Animal: no wall + small vacuole.
- Membrane selective + wall rigid support + audit. Ex: osmosis gate.

⚠ FATAL TRAP

Cell-only + 'plant asked' + wall-miss + gate-miss = wrong + fail! Wall + gate + audit.

★ NORCET HIT

HIT: Nucleus + mitochondria next — control + power + audit!

► Wall + gate + plant + guard + sort?

Cell + membrane + audit.

NUCLEUS + MITOCHONDRIA

Control + power + audit! DNA + ATP + audit — control + power + audit!

**Nucleus DNA +
 control + verify**

**Mitochondria ATP +
 power + audit**

**Control + power
 + pride!**

◆ POWER — RUNS AS

- Nucleus: DNA + control + verify. Ex: brain of cell.
- Mitochondria: ATP + powerhouse + aerobic + audit. Ex: muscle rich.

⚠ FATAL TRAP

Part-only + 'organelle asked' + DNA-miss + ATP-miss =
 wrong + fail! DNA + ATP + audit.

★ NORCET HIT

HIT: Ribosome + ER/Golgi next — protein + pack +
 audit!

► DNA + ATP + control + power + link?

Nucleus + mito + audit.

Protein + transport + pack + audit! Factory + road + post office + audit — make + pack + audit!

Ribosome protein +
factory + verify

ER road + Golgi
pack + audit

Make + pack
+ pride!

Σ FACTORY — SHIPS AS

- Ribosome: protein factory + verify. Ex: free + bound.
- ER transports + Golgi packs/ships + audit. Ex: post office.

⚠ FATAL TRAP

Organelle-only + 'cell asked' + make-miss + pack-miss
= wrong + fail! Make + pack + audit.

★ NORCET HIT

HIT: Till end: division + systems + audit!

► Make + pack + road + factory + link?

Ribosome + ER/Golgi + audit.

Suicide bags + kitchen + audit! Digest + photosynthesis + audit — digest + cook + audit!

**Lysosome/vacuole +
digest + verify**

**Chloroplast +
photosynthesis + audit**

**Digest + cook
+ pride!**

★ BAG — CLEANS AS

- Lysosome: suicide bags + digest + vacuole stores + verify. Ex: waste removal.
- Chloroplast: chlorophyll + photosynthesis + plants only + audit. Ex: green leaves.

⚠ **FATAL TRAP**

Green-only + 'cell asked' + digest-miss + pigment-miss
= wrong + fail! Digest + pigment + audit.

★ **NORCET HIT**

HIT: Mitosis + meiosis next — divide + halve + audit!

► **Digest + pigment + store + bag + sort?**

Lysosome + chloro + audit.

Equal divide vs halve + audit! Growth vs gametes + audit — copy + halve + audit!

Mitosis equal +
growth + verify

Meiosis halve +
gametes + audit

Copy + halve
+ pride!

▲ DIVIDE — SPLITS AS

- Mitosis: 1 into 2 equal + growth/repair + verify. Ex: skin healing.
- Meiosis: halve + gametes + variation + audit. Ex: sperm/ovum.

⚠ FATAL TRAP

Divide-only + 'division asked' + equal-miss +
halve-miss = wrong + fail! Equal + halve + audit.

★ NORCET HIT

HIT: Vs-table + digestive next — compare + gut + audit!

► Equal + halve + growth + gamete + sort?

Mitosis + meiosis + audit.

Compare + mouth to anus + audit! Number vs purpose + order + glands + audit — compare + order + audit!

Vs table +
compare + verify

Gut order + glands
+ enzymes + audit

Compare + order
+ pride!

◆ GUT — DIGESTS AS

- Vs: mitosis 2 equal vs meiosis 4 half + purpose + verify.
- Gut: mouth/foodpipe/stomach/intestine + liver/pancreas + audit. Ex: villi absorb.

⚠ FATAL TRAP

Order-only + 'gut asked' + compare-miss + gland-miss
= wrong + fail! Compare + gland + audit.

★ NORCET HIT

HIT: Respiratory + circulatory next — breathe + pump +
audit!

► Compare + order + gland + gut + check?

Vs + digestive + audit.

Breathe + pump + audit! Alveoli exchange + heart double + audit — exchange + pump + audit!

Lungs/alveoli +
 exchange + verify

Heart/blood +
 double + audit

Exchange + pump
 + pride!

Σ PUMP — MOVES AS

- Lungs: alveoli exchange O₂/CO₂ + diaphragm + verify. Ex: asthma narrows.
- Heart: double circulation + arteries/veins + RBC carry + audit. Ex: pulse check.

⚠ FATAL TRAP

Organ-only + 'system asked' + exchange-miss +
 circuit-miss = wrong + fail! Exchange + circuit + audit.

★ NORCET HIT

HIT: Nervous + endocrine next — signal + hormone +
 audit!

► Exchange + circuit + pump + breathe + link?

Lungs + heart + audit.

Fast signal vs slow hormone + audit! Neuron/reflex + glands + audit — fast vs slow + audit!

Neuron/reflex +
fast + verify

Glands/hormones +
slow + audit

Fast vs slow
+ pride!

★ SIGNAL — SENDS AS

- Nervous: neuron + brain/spinal + reflex fast + verify. Ex: knee jerk.
- Endocrine: pituitary/thyroid/insulin + slow + audit. Ex: diabetes insulin low.

⚠ FATAL TRAP

Gland-only + 'control asked' + speed-miss + gland-miss
= wrong + fail! Speed + gland + audit.

★ NORCET HIT

HIT: Excretory + largest next — filter + skin + audit!

► Fast + slow + gland + signal + sort?

Nervous + endocrine + audit.

Filter + skin/liver/bone + audit! Kidney nephron + biggest/smallest + audit — filter + biggest + audit!

**Kidney/nephron +
filter + verify**

**Skin/liver/femur +
biggest + audit**

**Filter + biggest
+ pride!**

▲ FILTER — TRAPS AS

- Kidney: nephron filters + urine + skin/lungs join + verify.
- Largest: skin organ + liver gland + femur bone + audit. Ex: classic traps.

⚠ FATAL TRAP

Kidney-only + 'largest asked' + unit-miss + trap-miss = wrong + fail! Unit + trap + audit.

★ NORCET HIT

HIT: Nutrients + water next — macro + fibre + audit!

► Filter + biggest + unit + trap + check?

Kidney + largest + audit.

Carb/protein/fat + water/fibre + audit! Energy/growth + roughage + audit — fuel + rough + audit!

**Carb/protein/fat +
roles + verify**

**Water/fibre +
roughage + audit**

**Fuel + rough
+ pride!**

◆ FUEL — FEEDS AS

- Macro: carbs energy + protein growth + fat reserve + verify. Ex: dal protein.
- Water solvent + fibre roughage + audit. Ex: salads + fluids.

⚠ FATAL TRAP

Food-only + 'nutrient asked' + role-miss + rough-miss =
wrong + fail! Role + rough + audit.

★ NORCET HIT

HIT: Vitamins A/B next — night + complex + audit!

► Fuel + rough + role + water + sort?

Macro + fibre + audit.

VITAMIN A + B COMPLEX

Night vision + complex set + audit! Retinol + B1/B12 + deficiency + audit — sight + set + audit!

**Vit A night/
retinol + verify**

**B1/beriberi + B12/
anaemia + audit**

**Sight + set
+ pride!**

Σ SIGHT — SEES AS

- A: night vision + retinol + xerophthalmia + verify. Ex: carrots + mango.
- B: B1 beriberi + B12 anaemia + whole set + audit. Ex: whole grains.

⚠ FATAL TRAP

Vitamin-only + 'A asked' + disease-miss + source-miss
= wrong + fail! Disease + source + audit.

★ NORCET HIT

HIT: C + D next — citrus + sun + audit!

► Sight + set + disease + source + link?

A + B + audit.

VITAMIN C + D

Citrus + sun + audit! Scurvy + rickets + audit — citrus + sun + audit!

**Vit C citrus/
scurvy + verify**

**Vit D sun/
rickets + audit**

**Citrus + sun
+ pride!**

★ SUN — BUILDS AS

- C: citrus/amlā + scurvy + heals + verify. Ex: lemon + guava.
- D: sun + rickets + bones + audit. Ex: milk + morning sun.

⚠ **FATAL TRAP**

Fruit-only + 'vitamin asked' + disease-miss +
source-miss = wrong + fail! Disease + source + audit.

★ **NORCET HIT**

HIT: E/K + solubility next — clot + fat/water + audit!

► **Citrus + sun + disease + bone + link?**

C + D + audit.

Clot + fat/water split + Hb + audit! K clots + A/D/E/K fat + B/C water + audit — clot + split + audit!

E/K + clot +
sources + verify

Fat vs water +
iron Hb + audit

Clot + split
+ pride!

▲ CLOT — SPLITS AS

- E protects + K clots + verify. Ex: greens + oils.
- Fat: A/D/E/K + Water: B/C + Iron Hb anaemia + audit. Ex: jaggery + greens.

⚠ FATAL TRAP

Letter-only + 'vitamin asked' + split-miss + iron-miss =
wrong + fail! Split + iron + audit.

★ NORCET HIT

HIT: Iodine + calcium next — goitre + bones + audit!

► Clot + split + Hb + group + sort?

E/K + split + iron + audit.

Goitre + bones + audit! Salt + milk + audit — gland + bone + audit!

**Iodine goitre +
salt + verify**

**Calcium bones +
milk + audit**

**Gland + bone
+ pride!**

◆ BONE — STRENGTHENS AS

- Iodine: thyroid + goitre + iodized salt + verify.
- Calcium: bones/teeth + milk/ragi + audit. Ex: osteoporosis low.

⚠ **FATAL TRAP**

Mineral-only + 'mineral asked' + disease-miss +
source-miss = wrong + fail! Disease + source + audit.

★ **NORCET HIT**

HIT: Till end: disease + plants + audit!

► **Gland + bone + salt + milk + link?**

Iodine + calcium + audit.

Spread vs lifestyle + germ types + audit! Virus/bacteria/fungus + examples + audit — spread + germ + audit!

Communicable vs
non + verify

Virus/bacteria/fungus
+ examples + audit

Spread + germ
+ pride!

★ SPREAD — SORTS AS

- Communicable spreads + noncommunicable lifestyle + verify. Ex: flu vs diabetes.
- Pathogens: virus/bacteria/fungus/protozoa + audit. Ex: TB bacteria.

⚠ FATAL TRAP

Germ-only + 'disease asked' + spread-miss + type-miss
= wrong + fail! Spread + type + audit.

★ NORCET HIT

HIT: Vectors + transmission next — carrier + route +
audit!

► Spread + germ + type + disease + sort?

Spread + pathogen + audit.

Carrier + air/water/food/contact + audit! Mosquito/fly + droplet/feco-oral + audit — carrier + route + audit!

Mosquito/fly +
carrier + verify

Air/water/food/
contact + audit

Carrier + route
+ pride!

▲ ROUTE — CARRIES AS

- Vectors: mosquito/fly/rat carry + verify. Ex: malaria + plague.
- Routes: air/water/food/contact + audit. Ex: ORS for watery loss.

⚠ FATAL TRAP

Bite-only + 'spreads asked' + carrier-miss + route-miss
= wrong + fail! Carrier + route + audit.

★ NORCET HIT

HIT: Immunity + antigen next — defend + flag + audit!

► Carrier + route + bite + disease + link?

Vector + route + audit.

Born vs built + flag vs fighter + audit! Innate/acquired + vaccine link + audit — born vs built + audit!

**Innate vs acquired
+ defend + verify**

**Antigen flag +
antibody fight + audit**

**Born vs built
+ pride!**

◆ DEFEND — FLAGS AS

- Innate born + acquired after expose/vaccine + verify. Ex: skin + memory.
- Antigen flags + antibody fights + specific + audit. Ex: blood groups.

⚠ FATAL TRAP

Immune-only + 'defence asked' + born-miss + flag-miss
= wrong + fail! Born + flag + audit.

★ NORCET HIT

HIT: Vaccine + antibiotics next — prevent +
bacteria-only + audit!

► Born + flag + fight + defend + sort?

Immunity + antigen + audit.

VACCINE + ANTIBIOTICS TRAP

Prevent + bacteria-only + audit! Weak/killed + no-virus + audit — prevent + limit + audit!

**Vaccine weak/killed
 + prevent + verify**

**Antibiotics bacterial/
 no-virus + audit**

**Prevent + limit
 + pride!**

Σ SHIELD — LIMITS AS

- Vaccine: weak/killed + trains + verify. Ex: BCG/polio + UIP link.
- Antibiotics kill bacteria + useless in virus + complete course + audit. Ex: no self-dose.

⚠ FATAL TRAP

Drug-only + 'cure asked' + prevent-miss + virus-miss =
 wrong + fail! Prevent + virus + audit.

★ NORCET HIT

HIT: Prevention + photosynthesis next — hygiene +
 sun-cook + audit!

► Prevent + limit + virus + shield + check?

Vaccine + antibiotic + audit.

Hygiene + sun-cook + audit! Clean hands + CO₂ + water + glucose + audit — hygiene + cook + audit!

Hygiene/vaccine +
prevent + verify

Sun/chlorophyll +
glucose + audit

Hygiene + cook
+ pride!

★ COOK — PREVENTS AS

- Prevention: hygiene/vaccine/vector + verify. Ex: handwash + nets.
- Photosynthesis: CO₂ + water + sun + chlorophyll = glucose + audit. Ex: green lab.

⚠ FATAL TRAP

Green-only + 'plant asked' + prevent-miss + equation-miss = wrong + fail! Prevent + equation + audit.

★ NORCET HIT

HIT: Factors + transpiration next — rate + loss + audit!

► Hygiene + equation + cook + prevent + link?

Prevent + photo + audit.

Rate levers + water loss + audit! Light/CO2/water + stomata + audit — rate + loss + audit!

Light/CO2/water +
rate + verify

Stomata loss +
pull + audit

Rate + loss
+ pride!

▲ LOSS — PULLS AS

- Factors: light/CO2/water/temp + rate + verify. Ex: cloudy slows.
- Transpiration: stomata loss + pulls water up + audit. Ex: wilting noon.

⚠ FATAL TRAP

Sun-only + 'plant asked' + factor-miss + loss-miss =
wrong + fail! Factor + loss + audit.

★ NORCET HIT

HIT: Transport + reproduction next — xylem + flower +
audit!

► Rate + loss + pull + plant + check?

Factors + transpire + audit.

PLANT TRANSPORT + SEXUAL REPRO

Xylem/phloem + flower/fruit + audit! Up-water + down-food + pollinate + audit — pipes + flower + audit!

**Xylem/phloem +
pipes + verify**

**Flower/fruit/seed +
pollinate + audit**

**Pipes + flower
+ pride!**

◆ PIPE — FLOWERS AS

- Xylem water up + phloem food both ways + verify. Ex: girdling test.
- Flower: pollinate + fertilize + fruit/seed + audit. Ex: mango seed.

⚠ **FATAL TRAP**

Pipe-only + 'plant asked' + direction-miss + flower-miss
= wrong + fail! Direction + flower + audit.

★ **NORCET HIT**

HIT: Asexual + auxin next — clone + bend + audit!

► **Pipes + flower + direction + seed + link?**

Transport + repro + audit.

ASEXUAL + AUXIN

Clone + bend-to-light + audit! Runner/bud/graft + tip + audit — clone + bend + audit!

Runner/bud/graft +
clone + verify

Auxin tip + bend
+ light + audit

Clone + bend
+ pride!

Σ CLONE — BENDS AS

- Asexual: runner/bud/graft/tissue + same + verify. Ex: potato + rose.
- Auxin: tip + stem bends to light + root grows + audit. Ex: sunflower tracks.

⚠ FATAL TRAP

Seed-only + 'repro asked' + clone-miss + bend-miss =
wrong + fail! Clone + bend + audit.

★ NORCET HIT

HIT: Other hormones + tropisms next — ripen + move +
audit!

► Clone + bend + tip + light + sort?

Asexual + auxin + audit.

HORMONE SET + TROPISMS

Ripen/sleep + move-to-stimulus + audit! Gibberellin/cytokinin/ABA/ethylene + photo/geo + audit — ripen + move + audit!

Gib/cyt/ABA/ethyl
+ roles + verify

Photo/geo/hydro +
moves + audit

Ripen + move
+ pride!

★ MOVE — RIPENS AS

- Gibberellin stem + cytokinin divide + ABA sleep + ethylene ripen + verify.
- Tropisms: photo/geo/hydro moves + audit. Ex: roots down + fruit ripen.

⚠ FATAL TRAP

Hormone-only + 'plant asked' + role-miss + move-miss
= wrong + fail! Role + move + audit.

★ NORCET HIT

HIT: Ecosystem + chains next — unit + links + audit!

► Role + move + ripen + hormone + link?

Hormones + tropism + audit.

Unit + links + audit! Producers/consumers + linear + audit — unit + links + audit!

**Ecosystem unit +
 parts + verify**

**Chain links +
 linear + audit**

**Unit + links
 + pride!**

▲ UNIT — LINKS AS

- Ecosystem: living + nonliving + interact + verify. Ex: pond + forest.
- Chain: linear links + grass/deer/lion + audit. Ex: who eats whom.

⚠ FATAL TRAP

Green-only + 'eco asked' + unit-miss + link-miss =
 wrong + fail! Unit + link + audit.

★ NORCET HIT

HIT: Web + trophic next — net + levels + audit!

► Unit + links + living + chain + sort?

Eco + chain + audit.

FOOD WEB + TROPHIC LEVELS

Net + levels + audit! Interlinked + 10 percent + audit — net + levels + audit!

Web net +
 interlinked + verify

Trophic + 10 pct
 + energy + audit

Net + levels
 + pride!

◆ NET — LEVELS AS

- Web: many chains net + stable + verify. Ex: forest net.
- Trophic: levels + 10 percent up + audit. Ex: less top predators.

⚠ FATAL TRAP

Chain-only + 'web asked' + net-miss + percent-miss =
 wrong + fail! Net + percent + audit.

★ NORCET HIT

HIT: Till end: pyramids + conservation + audit!

► Net + levels + percent + web + check?

Web + trophic + audit.

Shape + variety + audit! Number/biomass/energy + genes/species + audit — shape + variety + audit!

**Pyramids shape +
 types + verify**

**Biodiversity genes/
 species + audit**

**Shape + variety
 + pride!**

★ SHAPE — VARIES AS

- Pyramids: number/biomass/energy + upright/inverted + verify. Ex: tree pyramid.
- Biodiversity: genes/species/eco + hotspots + audit. Ex: Western Ghats.

⚠ FATAL TRAP

Shape-only + 'eco asked' + type-miss + variety-miss =
 wrong + fail! Type + variety + audit.

★ NORCET HIT

HIT: Conservation + air next — home vs away + smoke
 + audit!

► Shape + variety + type + hotspot + sort?

Pyramid + bio + audit.

Home vs away + smoke + audit! Sanctuary/zoo + SPM/RSPM + audit — home vs away + audit!

In-situ/ex-situ +
examples + verify

Air SPM/RSPM +
sources + audit

Home vs away
+ pride!

▲ HOME — SAVES AS

- In-situ home + ex-situ away + verify. Ex: Kaziranga vs zoo/seed bank.
- Air: SPM/RSPM + vehicles/burn + audit. Ex: Delhi smog + masks.

⚠ FATAL TRAP

Park-only + 'save asked' + home-miss + pollutant-miss
= wrong + fail! Home + pollutant + audit.

★ NORCET HIT

HIT: Water + greenhouse next — dirty + trap + audit!

► Home + pollutant + park + smoke + sort?

Conserve + air + audit.

WATER POLLUTION + GREENHOUSE EFFECT

Dirty + trap + audit! BOD/sewage + blanket + audit — dirty + blanket + audit!

**Water BOD/sewage
+ dirty + verify**

**Greenhouse blanket
+ trap + audit**

**Dirty + blanket
+ pride!**

◆ BLANKET — TRAPS AS

- Water: sewage/industry + BOD + verify. Ex: Ganga + Yamuna.
- Greenhouse: blanket + warms + natural + excess + audit. Ex: glasshouse.

⚠ **FATAL TRAP**

Dirty-only + 'pollute asked' + BOD-miss + blanket-miss
= wrong + fail! BOD + blanket + audit.

★ **NORCET HIT**

HIT: Gases + ozone next — list + shield + audit!

► **Dirty + blanket + BOD + warm + link?**

Water + green + audit.

CO2/CH4/N2O + UV block + audit! Exam list + CFC hole + audit — list + shield + audit!

CO2/CH4/N2O +
list + verify

Ozone UV + CFC
hole + audit

List + shield
+ pride!

Σ SHIELD — BLOCKS AS

- Gases: CO2/CH4/N2O/CFC + sources + verify. Ex: paddy + vehicles.
- Ozone blocks UV + CFC hole + Montreal + audit. Ex: sunscreen earth.

⚠ FATAL TRAP

CO2-only + 'gas asked' + list-miss + shield-miss =
wrong + fail! List + shield + audit.

★ NORCET HIT

HIT: Acid rain + waste next — sour + 3R + audit!

► List + shield + UV + hole + check?

Gases + ozone + audit.

ACID RAIN + WASTE HIERARCHY

Sour + 3R + audit! SO₂/NO_x + reduce/reuse/recycle + audit — sour + 3R + audit!

Acid SO₂/NO_x +
 sour + verify

3R + segregate
 + compost + audit

Sour + 3R
 + pride!

★ SOUR — SORTS AS

- Acid rain: SO₂/NO_x + marble damage + verify. Ex: Taj concern.
- Waste: reduce/reuse/recycle + segregate + audit. Ex: home bins.

⚠ FATAL TRAP

Rain-only + 'waste asked' + gas-miss + order-miss =
 wrong + fail! Gas + order + audit.

★ NORCET HIT

HIT: Scientists next — laws + names + audit!

► Sour + 3R + gas + order + sort?

Acid + waste + audit.

Laws + resistance + audit! Motion/gravity + $V = IR$ + audit — laws + ratio + audit!

Newton motion/
gravity + verify

Ohm $V=IR$ +
ratio + audit

Laws + ratio
+ pride!

▲ LAW — NAMES AS

- Newton: motion 3 + gravity + verify. Ex: apple + orbits.
- Ohm: $V = IR$ + resistance + audit. Ex: circuits.

⚠ FATAL TRAP

Name-only + 'scientist asked' + work-miss +
formula-miss = wrong + fail! Work + formula + audit.

★ NORCET HIT

HIT: Faraday + Rutherford next — induction + nucleus
+ audit!

► Work + formula + name + law + link?

Newton + Ohm + audit.

Induction + nucleus + audit! EMF + gold-foil + audit — induce + core + audit!

**Faraday induce/
EMF + verify**

**Rutherford foil +
nucleus + audit**

**Induce + core
+ pride!**

◆ CORE — INDUCES AS

- Faraday: induction + EMF + motor/generator + verify.
- Rutherford: gold foil + nucleus + audit. Ex: atom core.

⚠ FATAL TRAP

Name-only + 'scientist asked' + work-miss +
model-miss = wrong + fail! Work + model + audit.

★ NORCET HIT

HIT: Thomson + Chadwick next — electron + neutron +
audit!

► Induce + core + foil + EMF + link?

Faraday + Rutherford + audit.

Electron + neutron + audit! Plum pudding + discovery + audit — negative + neutral + audit!

Thomson electron
 + pudding + verify

Chadwick neutron
 + neutral + audit

Negative + neutral
 + pride!

Σ PARTICLE — FINDS AS

- Thomson: electron + plum pudding + verify. Ex: cathode rays.
- Chadwick: neutron + neutral + audit. Ex: completes trio.

⚠ FATAL TRAP

Name-only + 'scientist asked' + particle-miss +
 model-miss = wrong + fail! Particle + model + audit.

★ NORCET HIT

HIT: Mendeleev + Lavoisier next — table + oxygen +
 audit!

► Particle + model + negative + neutral + link?

Thomson + Chadwick + audit.

Table + oxygen + audit! Periodic + combustion + audit — table + air + audit!

**Mendeleev table
+ periodic + verify**

**Lavoisier oxygen +
conserve + audit**

**Table + air
+ pride!**

★ TABLE — BURNS AS

- Mendeleev: periodic table + gaps predicted + verify.
- Lavoisier: oxygen + combustion + mass conserved + audit. Ex: father of chemistry.

⚠ **FATAL TRAP**

Name-only + 'scientist asked' + work-miss +
predict-miss = wrong + fail! Work + predict + audit.

★ **NORCET HIT**

HIT: Pasteur + Jenner next — germs + vaccine + audit!

► **Table + air + predict + conserve + link?**

Mendeleev + Lavoisier + audit.

Germs + vaccine + audit! Pasteurize + smallpox + audit — germs + shield + audit!

Pasteur germs/
pasteurize + verify

Jenner smallpox/
vaccine + audit

Germs + shield
+ pride!

▲ GERM — SHIELDS AS

- Pasteur: germs + pasteurize + rabies + verify. Ex: milk safe.
- Jenner: smallpox vaccine + first + audit. Ex: UIP link.

⚠ FATAL TRAP

Name-only + 'scientist asked' + work-miss +
disease-miss = wrong + fail! Work + disease + audit.

★ NORCET HIT

HIT: Koch + Darwin next — postulates + evolve + audit!

► Germs + shield + milk + pox + link?

Pasteur + Jenner + audit.

Postulates + evolve + audit! Germ-proof + selection + audit — proof + select + audit!

Koch postulates +
TB + verify

Darwin selection +
evolve + audit

Proof + select
+ pride!

◆ PROOF — SELECTS AS

- Koch: postulates + TB/cholera + verify. Ex: germ proof rules.
- Darwin: natural selection + evolve + audit. Ex: finches + origin.

⚠ FATAL TRAP

Name-only + 'scientist asked' + proof-miss +
select-miss = wrong + fail! Proof + select + audit.

★ NORCET HIT

HIT: Book closed: 81pp + compendium + mocks +
audit!

► Proof + select + germ + evolve + close?

Koch + Darwin + audit.